

Jugla v404.02

Ernests Pētersons
Mārtiņš Vīksna
Rihards Putāns

Fortum

Watt's next? - Forecasting energy
consumption with AI

Modeling Techniques

For both 48-hour and 12-month period forecasts we tried using SARIMAX, which is algorithmic method based on statistical formulas, and DeepAR-style RNN model, which is in deep-learning based ML architecture.

Reasons why we chose SARIMAX:

- Via algorithm parameters it supports providing cycle / seasonal period lengths like year, week or day, which all are present in given data.
- It supports exogenous drivers, like prices, temperatures etc.
- Quick to implement, as it doesn't require much configuration and adaptation.

Potential problems with SARIMAX:

- Requires adaption for each group separately.
- It only supports linear relationships with exogenous drivers.
- Hard to scale with many exogenous drivers.
- Weak with long-term dependencies on exogenous drivers, like consumption depending on factors changing over many years, for example, policies or political events.

Reasons why we chose DeepAR-style RNN:

- Unlike SARIMAX, with this method we can learn nonlinear relationships.
- Unlike SARIMAX, we can also handle long-term dependencies.
- It supports knowledge sharing between all groups, learning shared patterns.

Potential problems with SARIMAX:

- Takes longer to implement and to train.
- With given data creates risks of overfitting.

Feature Selection & External Data

For both models we tried incorporating these data:

- Electricity prices, provided with the consumption data.
- Temperature, by the hour, averaged over longer periods(source: <https://open-meteo.com> API).
- Heating demand, by the hour, averaged over longer periods (source: <https://open-meteo.com> API).
- Data on productivity of the Finnish production industry (source: https://ec.europa.eu/eurostat/databrowser/view/sts_inpr_m/default/table?lang=en).
- Data on production in construction in Finland (source: https://ec.europa.eu/eurostat/databrowser/view/sts_copr_m/default/table?lang=en).

For SARIMAX in the end we used only electricity prices and heating demand, as this proved to be the best result producing data configuration out of available ones.

For DeepAR-style RNN we included all beforehand mentioned data.

Model Training & Validation

As validation we computed the MAPE metric alongside with other, to determine accuracy of our methods.

For both methods, in the 12-month forecast we took the first two years of data and forecasted the third year, then compared with the actual values. In the 48-hour forecast, we took 14 days of the data and forecasted the following two.

Business Understanding

By search and analysis of external data, and incorporation of it into our methods, we can make the predictions more precise and potentially adjusted on per-group basis.

Results Summary

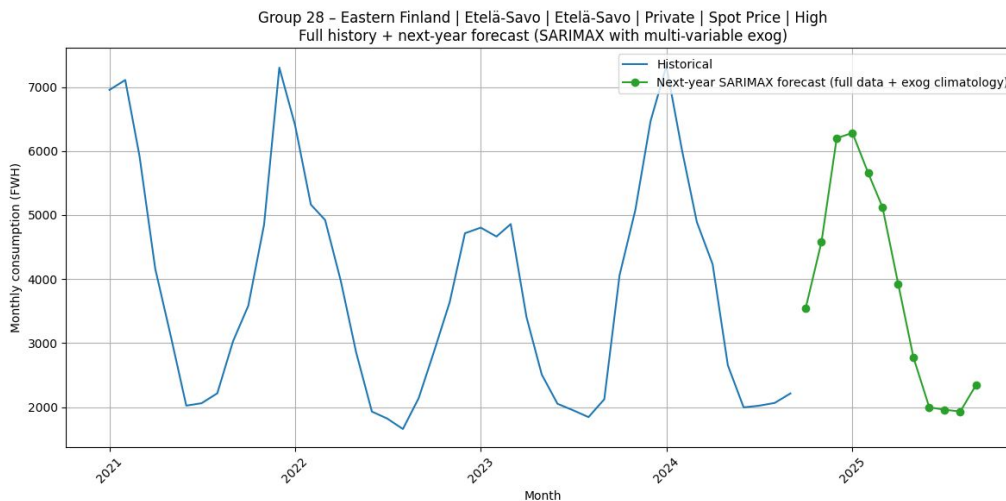
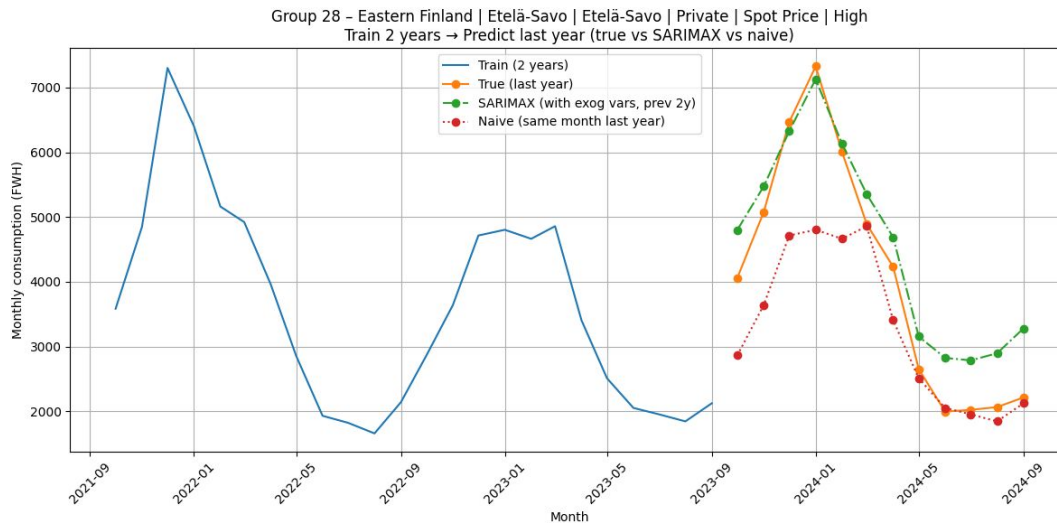
MAPE - Mean Absolute Percentage Error

Naive estimate - same data from previous week / year

12-month forecasts:

- SARIMAX:
 - MAPE: 9.38%

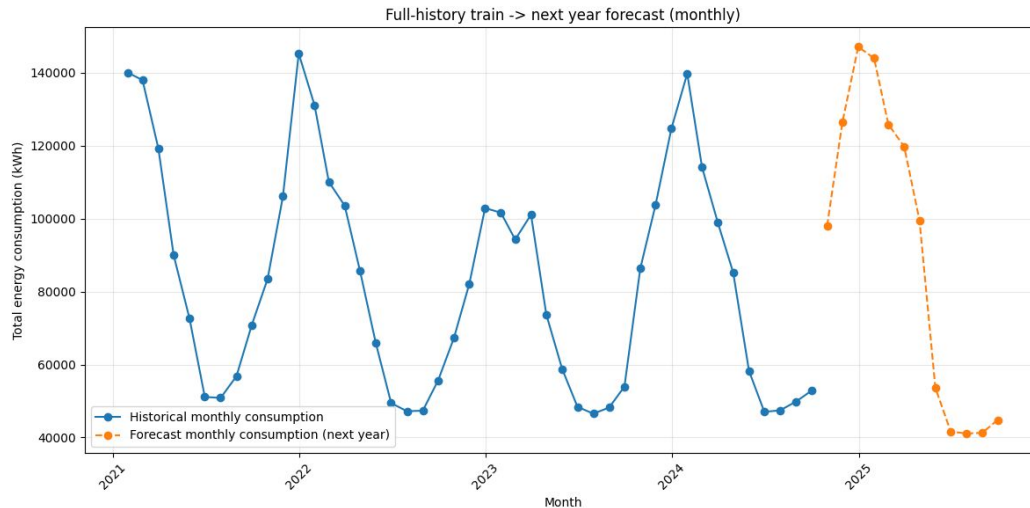
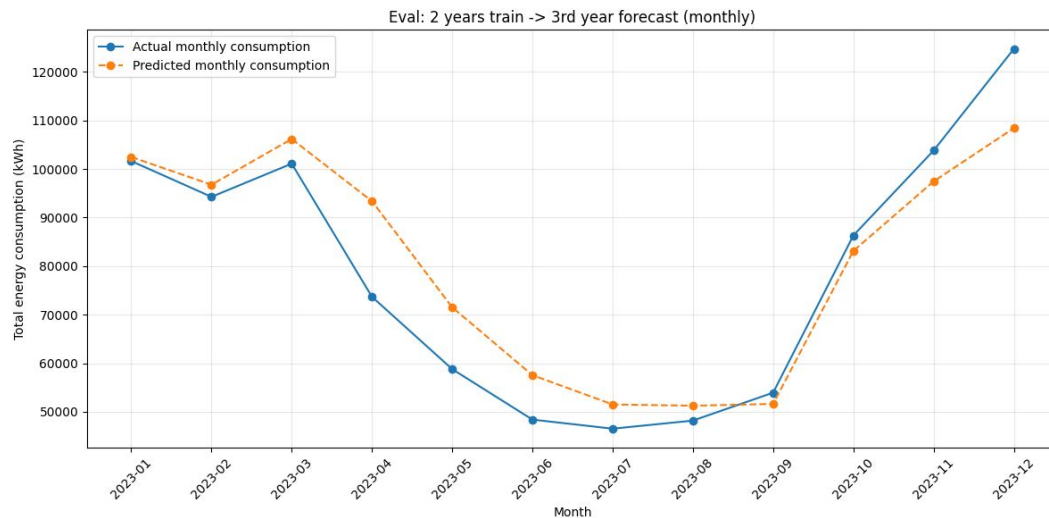
Naive: 10.21%



12-month forecasts:

- DeepAR-style RNN:
 - MAPE: 16.02%

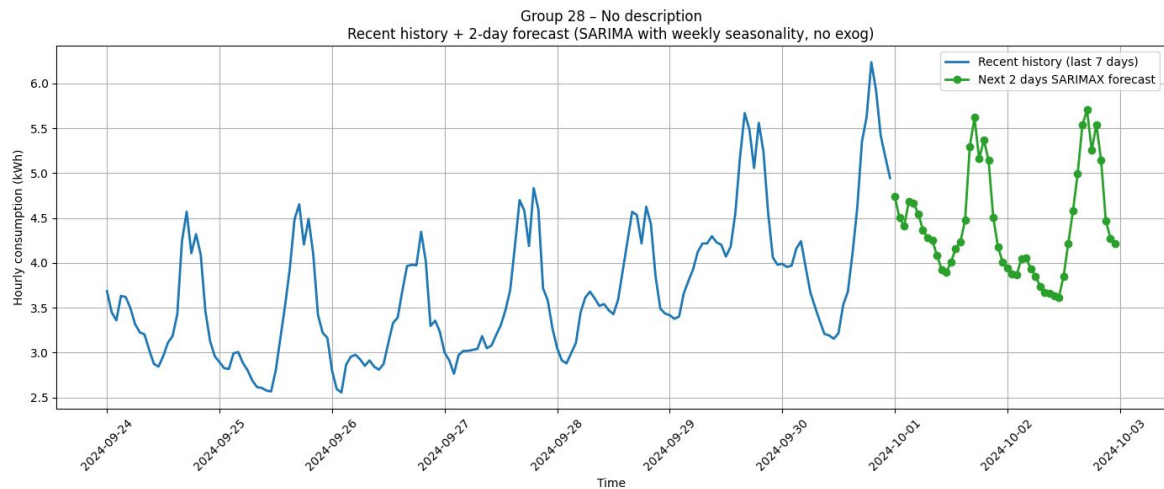
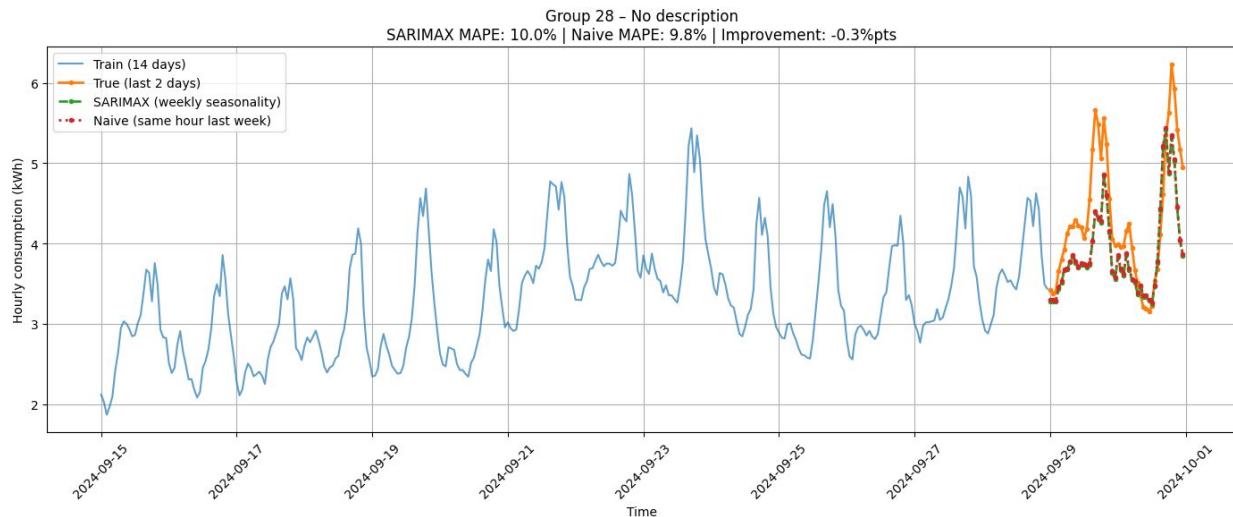
Naive: 10.21%



48-hour forecasts:

- SARIMAX:
 - MAPE: 7.33%

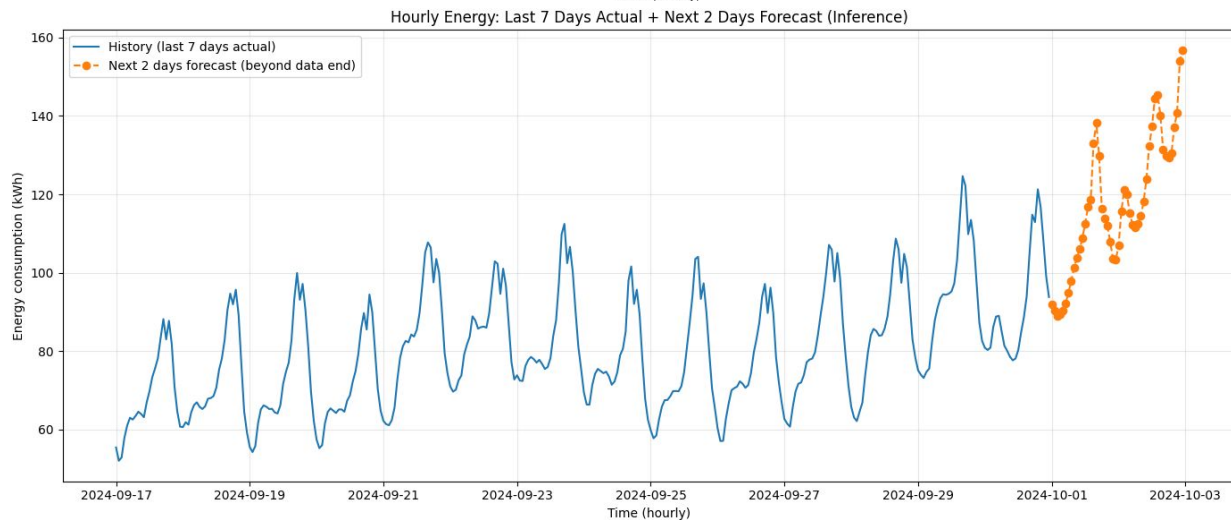
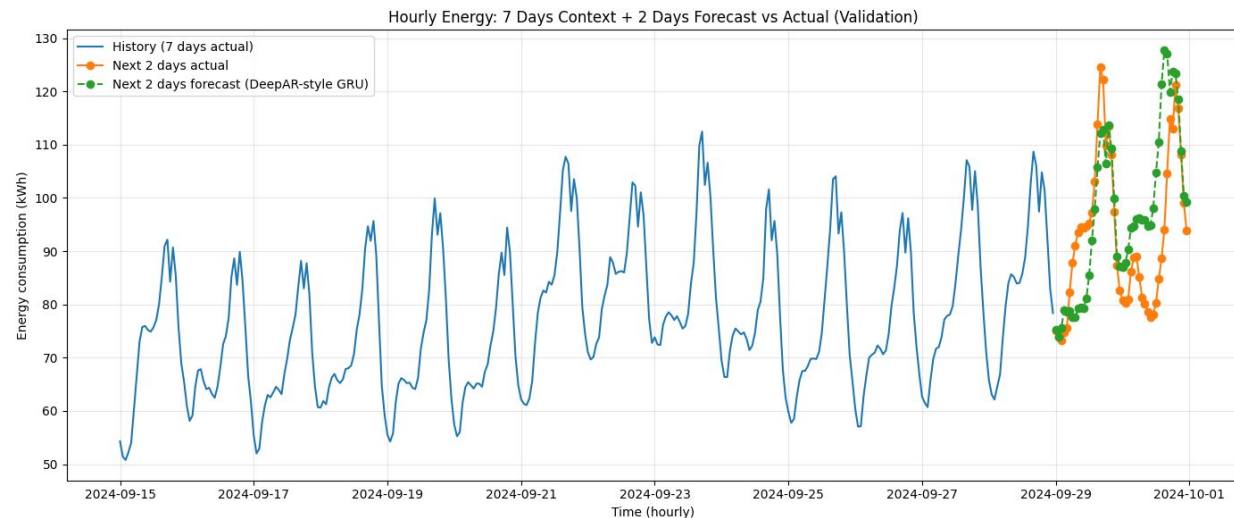
Naive: 7.62%



48-hour forecasts:

- DeepAR-style RNN:
 - MAPE: 10.75%

Naive: 7.62%



Find our code and full results here:
github.com/erpeetersons/Junction_Jugla_v404.02